Jaideepsinh Dabhi

Lead Research Engineer
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Summary

Lead Research Engineer with 5+ years of experience at the intersection of artificial intelligence, bioinformatics, and data science. Proven track record of taking ideas from concept to production, from agentic AI systems and LLM/VLM-powered platforms to RAG pipelines, digital pathology models, and cloud-native genomic workflows. Experienced in leading cross-functional teams and translating complex scientific and business challenges into scalable, high-impact solutions across precision oncology, healthcare, and government technology. Fluent in Python, PyTorch, and AWS, with a steady commitment to responsible AI and continuous innovation.

Work Experience

Genesilico — Delhi, India (Remote)
Lead Research Engineer, AI & Bioinformatics — Jan 2026 - Present

Leading the bioinformatics team in NGS and cancer research, spearheading **AI-driven pathology model** development and computational solutions for **precision oncology**.

- **Conceived and built PharmaOS from idea to production**, an AI coding agent that accelerates pharma R&D through automated cohort construction and in-silico clinical trial simulation, forecasting outcomes before costly real-world trials; delivered on an aggressive timeline to meet a critical business need and now the company's flagship product.
- **Spearheaded end-to-end Digital Twin reporting**, orchestrating the full journey from raw NGS data to client-ready reports for Sales, with zero operational disruption.
- **Directed NGS operations and pipeline engineering**: onboarded new panels, validated production workflows, and engineered a cloud-native somatic pipeline on AWS EKS (Nextflow/nf-core, Argo) that substantially reduced cost and turnaround time; mentored a team of bioinformaticians and built downstream tools translating genomic data into actionable drug recommendations (TMB, fusions, expression scoring).
- **Led AI engineering and R&D in digital pathology and radiology**, delivering models for H&E-to-mIF translation, tumor grading, and cancer cell detection, alongside spatial transcriptomics models that infer gene expression from H&E, backed by rigorous foundation-model benchmarking.
- **Deployed a VLM-powered inference platform** (Pathology Model) that empowers pathologists to interrogate large whole-slide images, and now spearheading a pan-cancer VLM that generates therapy-relevant insights directly from H&E slides.
- **Safeguarded data security and integrity** by managing on-prem infrastructure, and championed AI strategy by presenting roadmaps and technical vision to executive leadership.

KPMG India — Gandhinagar, India
AI/ML Consultant — Jun 2025 - Dec 2025

AI Consultant for Department of Science and Technology (DST), Gujarat Government, as part of GovTech division in KPMG Advisory.

- **Built and launched G-SAAR link (Smart AI Archive & Retrieval)**, an AI-powered search portal (Internal Government portal) digitising 30,000+ government circulars, rules, and resolutions for instant retrieval and faster policy decisions; launched by the Chief Minister of Gujarat on Good Governance Day.
- **Led the development and deployment of the Gujarat Government AI Stack** (aiservices.gujarat.gov.in), a transparent, citizen-centric platform unveiled by the Chief Minister at the Regional AI Summit.
- **Advised the AI CoE steering committee** on vendor POC procurement and rapid due diligence of AI proposals, ensuring responsible AI adoption without compromising citizen data security.
- **Shaped Gujarat's AI infrastructure strategy**, leading local GPU procurement and the compute needs analysis and roadmap.
- **Led development of a national-level MoHA analytics dashboard** tracking women and child welfare KPIs from pregnancy through adolescence, with LLM-driven email alerts that flag scheme-execution gaps to strengthen accountability.
- **Engineered a secure, end-to-end AI system for government form submission**, automating data extraction, form filling, and document verification to cut manual effort and widen citizen access.
- **Served on the AI Centre of Excellence (CoE) at GIFT City**, building Gujarati LLM and RAG tools for multiple departments, and guiding departments from use-case identification through POC to implementation.

HolboxAI — Ahmedabad, India
Data Scientist 2 — Aug 2024 - May 2025

- Successfully **Design - Develop - Execute** Various **POCs** and Solutions for Clients using **LLM, RAG, DL/ML.**
- Created Bioinformatics POCs and Solutions to streamline from **raw data to reporting** using **LLMs**.
- Developed **breast cancer segmentation models** using nnU-Net and a Transformer-based UNet (TransUNet) on multiphase MRI scans, achieving an average **IoU of 0.81**. Integrated results into a 3D viewer.
- Conducted research on histopathology analysis using various **foundational models** and collaborated with domain experts to gather feedback and continuously improve model performance.
- Developed a forecasting system using **SARIMAX** to predict user-preferred temperatures, and created apps, to optimize temperature settings for enhanced comfort and sleep quality and save energy.
- Developed a Security **Auditing System using LLMs** to analyze receipt images, extract relevant information for fraud detection, and identify anomalous transaction patterns for enhanced security monitoring.

Strand Life Sciences (Reliance) — Bengaluru, India (Remote)
Associate Bioinformatics Engineer II (Clinical Data Science) — Jan 2023 - Aug 2024

- Handled **clinical whole exome sequencing** panel and did variant detection and verification.
- Worked on **Machine Learning** techniques to simulate WES and develop tools for CNV detection with better results.
- Developed an end-to-end bioinformatics pipeline to process w**hole exome sequencing** (WES) data for **PGx** (GHI) report generation, integrating variant calling, annotation, and copy number variation (CNV) analysis for comprehensive patient profiling.
- **Led a team** to streamline the transition from one system to another without changing the current process.
- **Developed applications** for QC analysis and anomaly detection in sequencing samples.
- Developed a new pipeline for whole exome sequencing (WES) for SNPs and CNVs from **Dragen Enrichment** data.

NielsenIQ — Vadodara, India
Data Processing Specialist (Statistical Operations) — Jan 2021 - Feb 2022

- Led the design of the logic for the PER Automation project and drove it to successful delivery.
- Automated final market sign-off, saving the team 7–10 days with 0% quality escape.

- Actively participated in and contributed to the internal **Data Science project** initiatives. And executed **Statistical Estimation** &, provided root cause analysis of data validation and resolved issues at the Individual level.

Projects

JustDataML — Dec 2023 - May 2024
Self
- Simplified Machine Learning for Everyone
- A user-friendly tool designed to make machine learning accessible to all, regardless of technical background. JustDataML automates model selection, data preprocessing, and prediction, so you can focus on insights rather than coding complexity.

End-End Bioinformatics Pipeline for WES — May 2023 - Aug 2023
Strand Life Sciences
- Developed an end-to-end bioinformatics pipeline from Dragen Enrichment Data (Illumina Platform).
- It can detect CNVs and SNPs and filter variants that are relevant.

Prediction of DDoS Attack Using Machine Learning, Bengaluru, India — Oct 2022 - Nov 2022
IIIT-B
- ML Project for my Machine Learning Class in IIIT-B campus.
- In this project task was to handle Big Data and predict DDoS attack on Network
- Implemented big data processing techniques to analyze network traffic patterns and identify malicious activities
- Achieved high accuracy in threat classification contributing to enhanced cybersecurity posture

Common Pathways for Autism and Alzheimer's disease for Early diagnosis Using Machine Learning — Aug 2022 - Jan 2023
IBAB
- Using ML, we tried to Find Common Pathways and molecular processes to detect AD if a patient has ASD in Younger age.

Education

International Institute of Information Technology Bangalore, Bengaluru, India — Feb 2022 - Jan 2023
Post Graduate Diploma in Big Data Biology — GPA: 8.45

Gujarat University, Ahmedabad — May 2019 - May 2021
MSc in Statistics — GPA: 8.53

Gujarat University, Ahmedabad — Aug 2016 - May 2019
BSc in Statistics — GPA: 6.46

Skills

Top Skills: AI Agent, LLM, Python (Programming Language), AI/ML, Data Science, Cloud
DS/ML Skills: Statistics, Deep Learning, Computer Vision, Hypothesis Testing, Machine Learning Algorithms, Anomaly Detection, Random Forest, GenAI, LLMs, Data Visualization, Model Development, Vector DB
AWS: EC2, SageMaker, Bedrock, Lambda, Amazon S3, Boto3, AWS Glue
Language: Python, Linux, R Programming Language
Frameworks: Pandas, Scikit-Learn, Matplotlib, Seaborn, Pyspark, PyTorch, OpenCV, LangChain, SciPy, NumPy
Platforms: Jupyter Notebook, Visual Studio Code, Linux, AWS, Linux Server, Azure
Tools: Microsoft Office, Power BI, MySQL, Hadoop, Git, gradio, SAS
Soft Skills: People Management, Leadership, Take Ownership, Process Orientation, Communication, Problem Solving

Certifications

Machine Learning Specialty (MLS-C01), AWS — Jan 2025

AI Practitioner, AWS — Mar 2025

Foundations of Data Science, PadhAI - One-Fourth Labs (IIT Madras) — Sep 2021

Statistics with SAS, Coursera - SAS — Jul 2020

Data Science and Machine Learning, Fingertips Data Solutions — Nov 2020

Spatial Data Science: The New Frontier in Analytics, ESRI — Dec 2020